

we will place *DC Simulation* block, which can be found in the *Components* tab in the *Simulations* category.

To get the voltage between the two resistors (the divided voltage) as an output in the dataset after simulation, you need to label the wire by double clicking the wire and giving an appropriate name, or by using *ctrl + L* shortcut. Also, you can choose *Insert*→*Wire Label* menu entry. Fig. 6 shows how the final circuit would look like after following the above steps.

Simulation

Select *Simulation*→*Simulate* menu entry to start the simulation. Alternatively, you can click the simulation button (gearwheel) or press *F2* shortcut.

After the simulation, *Components* tab is changed to *diagrams* category and related data is displayed. Now you can choose the *Tabular* from *diagrams* and place it on the data display page. After dropping the *Tabular*, the diagram, a dialog box appears as shown in Fig. 7. Double click *divided.V* to add the graph (i.e., values in a *Tabular* plot) to the diagram. Similarly, other items listed in the dataset list can be put into the graph.

Now you can see the value of the node voltage *divided.V*, which is 0.5V. That was expected since the values of the resistors are the same and the DC voltage source produces 1V (Fig. 8).

Other features

Having made your first successful

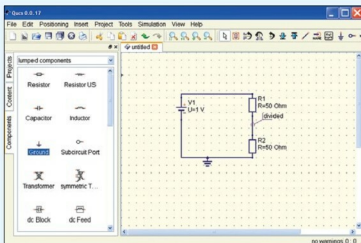


Fig. 6: The final circuit

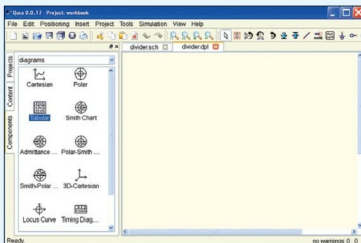


Fig. 7: A window displaying the diagrams category

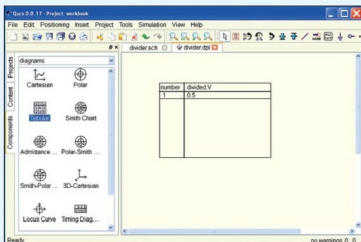


Fig. 8: A window displaying *divided.V* button

simulation of QUCS, now you can extend your simulation for a better understanding of the circuit by changing the component properties either

by clicking on the *divider.sch* tab or using the *F4* shortcut, or by selecting the *Simulation*→*View Data Display/Schematic* menu entry. Afterwards, double click on any resistor (say *R1*), which will open a component property dialog box where you can edit all the properties of the selected component. Change the values and properties of the component and observe the results.

Also, if you do not want QUCS to change automatically to the associated display data, you can change the behaviour of the document by changing the settings. You can go to the *document settings* dialog box by right clicking on the free space on the schematic area and selecting the *Document Settings* menu, item in the context menu which pops up or by selecting the *File*→*Document Settings* menu entry.

Useful tool

QUCS is an easy-to-use and handy tool for circuit designers. Not only it supports all kinds of simulation types but it also allows you to have multiple simulations with one design, by allowing you to change the parameters and properties of the components. QUCS helps in better understanding of the circuit design as well as components. Availability of mathematical equations makes it a useful tool for the designers and educators as well as

learners. ●

The author is a technical journalist at EFY Gurgaon